

SCHEMA	DETAILS	PREVIEW	TABLE EXPLORER	PREVIEW	INSIGHTS	LINEAGE	DATA PROFILE	DATA QUALITY
Row	name	gender	count					
1	Emma	F	20799					
2	Olivia	F	19674					
3	Sophia	F	18490					
4	Isabella	F	16950					
5	Ava	F	15586					
6	Mia	F	13442					
7	Emily	F	12562					
8	Abigail	F	11985					
9	Madison	F	10247					
10	Charlotte	F	10048					
11	Harper	F	9564					
12	Sofia	F	9542					
13	Avery	F	9517					
14	Elizabeth	F	9492					
15	Amelia	F	8727					
16	Evelyn	F	8692					
17	Ella	F	8489					
18	Chloe	F	8469					
19	Victoria	F	7955					
20	Aubrey	F	7589					
21	Grace	F	7554					
22	Zoey	F	7358					
23	Natalie	F	7061					
24	Addison	F	6950					
25	Lillian	F	6869					
26	Brooklyn	F	6767					
27	Lily	F	6727					
28	Hannah	F	6512					
29	Layla	F	6428					
30	Scarlett	F	5965					
31	Aria	F	5893					
32	Zoe	F	5828					
33	Samantha	F	5680					
34	Anna	F	5639					
35	Leah	F	5563					
36	Audrey	F	5531					
37	Ariana	F	5461					
38	Allison	F	5440					
39	Savannah	F	5433					

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Job history

"job_native_table" created. GO TO TABLE X

Open Editor | | | | |

- Screenshot the query results and include it in your lab notebook

Query results

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	name ▾	count ▾				
1	Emma	20799				
2	Olivia	19674				
3	Sophia	18490				
4	Isabella	16950				
5	Ava	15586				
6	Mia	13442				
7	Emily	12562				
8	Abigail	11985				
9	Madison	10247				
10	Charlotte	10048				
11	Harper	9564				
12	Sofia	9542				
13	Avery	9517				
14	Elizabeth	9492				
15	Amelia	8727				
16	Evelyn	8692				
17	Ella	8489				
18	Chloe	8469				
19	Victoria	7955				
20	Aubrey	7589				

```
jgafron@cloudshell:~ (cloud-gafron-jgafron)$ bq query "SELECT name, count FROM [cloud-gafron-jgafron.yob.yob_native_table] WHERE gender='M' ORDER BY count ASC LIMIT 10"
+-----+-----+
| name | count |
+-----+-----+
| Aari | 5 |
| Aaliyah | 5 |
| Aadian | 5 |
| Aaroh | 5 |
| Aarit | 5 |
| Aadiv | 5 |
| Aadhi | 5 |
| Aarohan | 5 |
| Aariyan | 5 |
| Aamer | 5 |
+-----+-----+
jgafron@cloudshell:~ (cloud-gafron-jgafron)$
```

```
cloud-gafron-jgafron> SELECT name, count FROM [cloud-gafron-jgafron.yob.yob_native_table]
+-----+-----+
| name | count |
+-----+-----+
| Noah | 19144 |
| Liam | 18342 |
| Mason | 17092 |
| Jacob | 16712 |
| William | 16687 |
| Ethan | 15619 |
| Michael | 15323 |
| Alexander | 15293 |
| James | 14301 |
| Daniel | 13829 |
+-----+-----+
```

```
+-----+-----+
cloud-gafron-jgafron> SELECT name, count FROM [cloud-gafron
+-----+-----+
| name | count |
+-----+-----+
| Joseph | 11995 |
+-----+-----+
cloud-gafron-jgafron> 
```

- Screenshot the query results and include it in your lab notebook

Untitled query

RUNSAVEDOWNLOADSHARESCHEDULEOPEN INMORE

```
1 SELECT name, count
2 FROM 'cloud-gafron-jgafron.yob.yob_biglake_table'
3 WHERE gender = 'F'
```

Query results

JOB INFORMATIONRESULTSCHARTJSONEXECUTION DETAILSEXECUTION GRAPH

Metadata caching is disabled. You can accelerate queries over external tables by enabling metadata caching. [Learn more.](#)

Row	name	count
1	Aarshi	5
2	Aaniylah	5
3	Aaryah	5
4	Aashirya	5
5	Aalimah	5
6	Aarielle	5
7	Aarabella	5
8	Aayra	5
9	Aarti	5
10	Aavya	5
11	Aashni	5
12	Aadrika	5
13	Aamyah	5
14	Aamilah	5
15	Abagael	5
16	Aayusha	5
17	Aarion	5
18	Aania	5
19	Aaiza	5
20	Aabriella	5

- **How much less data does this query process compare to the size of the table?**

It went from 21 GBs to 3 GBs, so about 18 GBs.

- **How many twins were born during this time range?**

375000

- **How much lighter on average are they compared to single babies?**

About 2 pounds less (From 7 pounds to 5 pounds)

- **Show the plots generated for the two most important features for your lab notebook**

Most definitely gestation weeks, but i would say gender is the second strongest predictor

- **What day saw the largest spike in trips to grocery and pharmacy stores?**

March 13th

- **On the day the stay-at-home order took effect (3/23/2020), what was the total impact on workplace trips?** On the day the stay-at-home order took effect (3/23/2020), what was the total impact on workplace trips?
- -49%

- **Which three airports were impacted the most in April 2020 (the month when lockdowns became widespread)?**

Detroit, McCarran, and San Francisco

- **Run the query again using the month of August 2020. Which three airports were impacted the most?**

Detroit, McCarran, and San Francisco

- **What table and columns identify the place name, the starting date, and the number of excess deaths from COVID-19?**

Table = excess_deaths, Place Name Column = placename, Date Column = start_date, Excess deaths column = excess_deaths

- **What table and columns identify the date, county, and deaths from COVID-19?**

Table = us_counties, Date column = date, County column = county, Deaths column = deaths

- **What table and columns identify the date, state, and confirmed cases of COVID-19?**

Table = Us_states, Date Column: date , State Column: state_name , Confirmed Cases Column: confirmed_cases

- **What table and columns identify a county code and the percentage of its residents that report they always wear masks?**

Table = Mask_use_by_county, County Code Column:county_fips_code, Percentage of Residents: always

- **Show a screenshot of the plot and the code used to generate it for your lab notebook**

```

|: from google.cloud import bigquery

client = bigquery.Client()

query_string = """
SELECT
    date,
    confirmed_cases
FROM
    `bigquery-public-data.covid19_nyt.us_states`
WHERE
    state_name = 'Oregon'
ORDER BY
    date ASC
"""

df = client.query(query_string).to_dataframe()

```

```

|:

```

	date	confirmed_cases
0	2020-02-28	1
1	2020-02-29	1
2	2020-03-01	2
3	2020-03-02	2
4	2020-03-03	2

```

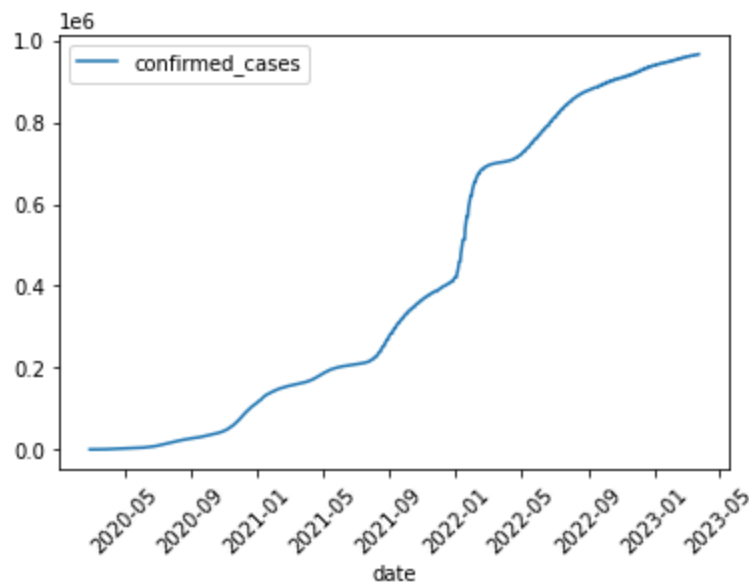
|: df.plot(x='date', y='confirmed_cases', kind='line', rot=45)

```

```

|: <matplotlib.axes._subplots.AxesSubplot at 0x7fd8249540d0>

```



- From within your Jupyter notebook, run the query and write code that shows the first 10 states that reached 1000 deaths from COVID-19. Take a screenshot for your lab notebook.


```
[18]: from google.cloud import bigquery
import pandas as pd

client = bigquery.Client()

query_string = """
SELECT
    state_name,
    MIN(date) AS date_of_1000
FROM
    `bigquery-public-data.covid19_nyt.us_states`
WHERE
    deaths > 1000
GROUP BY
    state_name
ORDER BY
    date_of_1000 ASC
LIMIT 10
"""

df = client.query(query_string).to_dataframe()

print(df)

import IPython.display as display
display.display(df)
```

```

      state_name  date_of_1000
0      New York    2020-03-29
1    New Jersey    2020-04-06
2      Michigan    2020-04-09
3    Louisiana    2020-04-14
4  Massachusetts    2020-04-15
5      Illinois    2020-04-16
6  Pennsylvania    2020-04-17
7      California    2020-04-17
8    Connecticut    2020-04-17
9        Florida    2020-04-24
```

	state_name	date_of_1000
0	New York	2020-03-29
1	New Jersey	2020-04-06
2	Michigan	2020-04-09
3	Louisiana	2020-04-14
4	Massachusetts	2020-04-15
5	Illinois	2020-04-16
6	Pennsylvania	2020-04-17
7	California	2020-04-17
8	Connecticut	2020-04-17
9	Florida	2020-04-24

- Take a screenshot for your lab notebook of the Top 5 counties and the states they are

```
client = bigquery.Client()

query_string = """
SELECT DISTINCT
    mu.county_fips_code,
    mu.always,
    ct.county,
    ct.state_name
FROM
    `bigquery-public-data.covid19_nyt.mask_use_by_county` AS mu
LEFT JOIN
    `bigquery-public-data.covid19_nyt.us_counties` AS ct
ON
    mu.county_fips_code = ct.county_fips_code
ORDER BY
    mu.always DESC
LIMIT 5
"""

df = client.query(query_string).to_dataframe()

display(df)

print(df)
```

	county_fips_code	always	county	state_name
0	06027	0.889	Inyo	California
1	36123	0.884	Yates	New York
2	48229	0.880	Hudspeth	Texas
3	06051	0.880	Mono	California
4	48141	0.877	El Paso	Texas

	county_fips_code	always	county	state_name
0	06027	0.889	Inyo	California
1	36123	0.884	Yates	New York
2	48229	0.880	Hudspeth	Texas
3	06051	0.880	Mono	California
4	48141	0.877	El Paso	Texas

located in.

- Plot the results and take a screenshot for your lab notebook.

```
client = bigquery.Client()

query_string = '''SELECT
    date,
    deaths
FROM
    `bigquery-public-data.covid19_nyt.us_counties`
WHERE
    county = 'Multnomah'
    AND state_name = 'Oregon'
ORDER BY
    date ASC;
'''

df = client.query(query_string).to_dataframe()

display(df)

print(df)
```

	date	deaths
0	2020-03-10	0
1	2020-03-11	0
2	2020-03-12	0
3	2020-03-13	0
4	2020-03-14	1
...	...	...
1022	2022-12-27	1426
1023	2022-12-28	1427
1024	2022-12-29	1427
1025	2022-12-30	1427
1026	2022-12-31	1427

1027 rows x 2 columns

	date	deaths
0	2020-03-10	0
1	2020-03-11	0
2	2020-03-12	0
3	2020-03-13	0
4	2020-03-14	1
...	...	...
1022	2022-12-27	1426
1023	2022-12-28	1427
1024	2022-12-29	1427
1025	2022-12-30	1427
1026	2022-12-31	1427

[1027 rows x 2 columns]

- Plot the results and take a screenshot for your lab notebook.

```
[23]: from google.cloud import bigquery
import pandas as pd
from IPython.display import display

client = bigquery.Client()

query_string = '''SELECT
    date,
    SUM(deaths) AS total_deaths
FROM
    `bigquery-public-data.covid19_nyt.us_counties`
WHERE
    state_name = 'Oregon'
GROUP BY
    date
ORDER BY
    date ASC;
'''

df = client.query(query_string).to_dataframe()

display(df)

print(df)
```

	date	total_deaths
0	2020-02-28	0
1	2020-02-29	0
2	2020-03-01	0
3	2020-03-02	0
4	2020-03-03	0
...	...	...
1033	2022-12-27	8953
1034	2022-12-28	8967
1035	2022-12-29	8967
1036	2022-12-30	8967
1037	2022-12-31	8967

1038 rows × 2 columns

	date	total_deaths
0	2020-02-28	0
1	2020-02-29	0
2	2020-03-01	0
3	2020-03-02	0
4	2020-03-03	0
...	...	...
1033	2022-12-27	8953
1034	2022-12-28	8967
1035	2022-12-29	8967
1036	2022-12-30	8967
1037	2022-12-31	8967

- How long did the job take to execute?

About 1 minute

- Examine `output.txt` and show the estimate of π calculated.

3.1416241514162415

```
42121931_0001.inprogress [CONTEXT ratelin  
Pi is roughly 3.1416241514162415  
25/03/06 06:25:38 INFO RequestTracker: De  
1242121931_0001.inprogress.2.058f0b79-d7c
```

- How long did the job take to execute? How much faster did it take?

About 30 Seconds, so about twice as fast

- How long did the job take to execute? How much faster did it take?

3.141439111414391

- **Where is the input taken from by default?**

`parser.add_argument(--input...)`

- **Where does the output go by default?**

`/tmp/output`

- **Examine both the `getPackages()` function and the `splitPackageName()` function. What operation does the `'PackageUse()'` transform implement?**

`PackageUse` extracts and counts package imports at each hierarchical level

- **Look up Beam's `CombinePerKey`. What operation does the `TotalUse` operation implement?**

It groups the package tuples by package name, then sums the counts for each package

- **Which operations correspond to a "Map"?**

`"GetImports" >> beam.FlatMap...`

`'PackageUse' >> beam.Flatmap...`

- **Which operation corresponds to a "Shuffle-Reduce"?**

`'TotalUse' >> beam.CombinePerKey...`

- **Which operation corresponds to a "Reduce"?**

`'TotalUse' >> beam.CombinePerKey(sum)`

`'Top_5' >> beam.transforms.combiners.Top.Of...`

- **Take a screenshot of its contents**
- **Explain what the data in this output file corresponds to based on your understanding of the program.**

```
jgafron@cloudshell:/tmp (cloud-gafron-jgafron)$ cat output-00000-of-00001
[('org', 45), ('org.apache', 44), ('org.apache.beam', 44), ('org.apache.beam.sdk', 43), ('org.apache.beam.sdk.transforms', 16)]
jgafron@cloudshell:/tmp (cloud-gafron-jgafron)$
```

To my understand, this is the top 5 most common Java packages out of all the .java files

- **What are the names of the stages in the pipeline?**

Read, split, pairwithone, groupandsum, format and write

- **Describe what each stage does.**

Read - Reads text from an input file, split - splits lines into words, Pairwithone - pairs words with a count of 1, groupandsum - groups and sums the word counts, format - formats the results as readable text, write - writes the final counts to a file

- Use `wc` with an appropriate flag to determine the number of different words in King Lear.

4784

- Use sort with appropriate flags to perform a *numeric* sort on the *key field* containing the count for each word in *descending* order. Pipe the output into `head` to show the top 3 words in King Lear and the number of times they appear

The - 786, I - 622 , And - 594

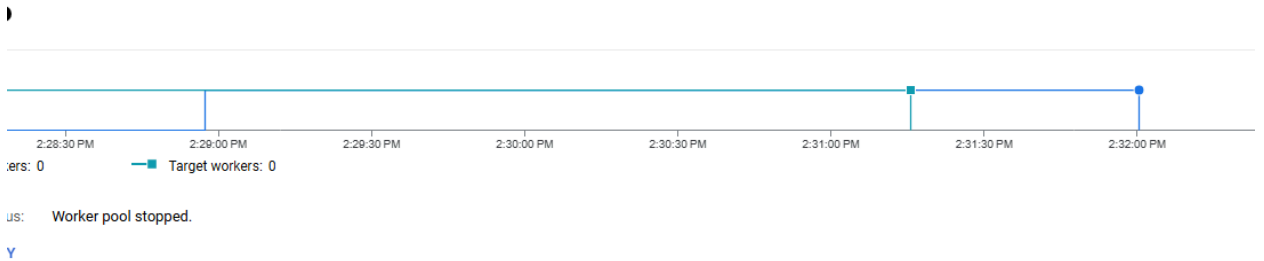
- Use the previous method to show the top 3 words in King Lear, case-insensitive, and the number of times they appear.

```
INFO:apache_beam.runners.worker.statecache:Created
INFO:apache_beam.io.filebasedsink:Starting final write
INFO:apache_beam.io.filebasedsink:Renamed 1 shard
(env) jgafron@cloudshell:~/training-data-analysis
the: 908
and: 738
i: 622
(env) jgafron@cloudshell:~/training-data-analysis
```

- The part of the job graph that has taken the longest time to complete.

Write Succeeded

- The autoscaling graph showing when the worker was created and stopped.



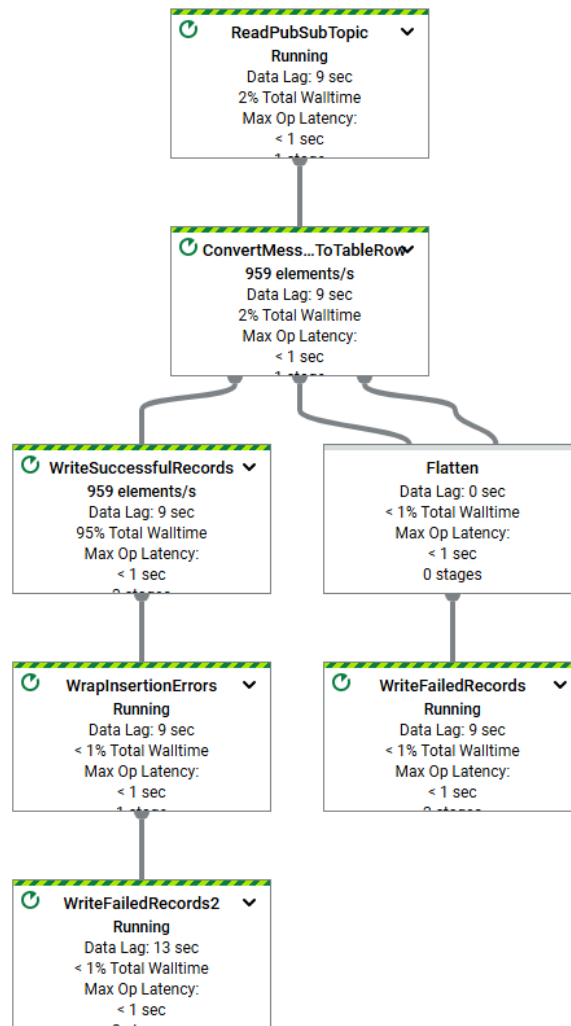
- Examine the output directory in Cloud Storage. How many files has the final write stage in the pipeline created?

1 File

- Take a screenshot listing the different fields of this object.


```
Created subscription [projects/cloud-gafron-jgafron/subscriptions/taxis]
(env) jgafron@cloudshell:~ (cloud-gafron-jgafron)$ gcloud pubsub subscri
DATA: {"ride_id":"93d07352-8c1c-460f-9b4e-738cf448af28","point_idx":403
MESSAGE_ID: 13744925078390288
ORDERING_KEY:
ATTRIBUTES: ts=2025-03-06T17:40:14.17258-05:00
DELIVERY_ATTEMPT:
ACK_STATUS: SUCCESS
(env) jgafron@cloudshell:~ (cloud-gafron-jgafron)$
```

- Take a screenshot of the pipeline that includes its stages and the number of elements per second being handled by individual stages.



Take a screenshot showing the number of passengers and the amount paid for the first ride

•

```
1 SELECT ride_id, passenger_count, meter_reading
2 FROM `cloud-gafron-jgafron.taxirides.realtime`
3 ORDER BY timestamp ASC
4 LIMIT 1;
```

Query results

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	ride_id	passenger_count	meter_reading			
1	7c360b89-d6b9-4f9f-b232-9ae...	1	30.569109			

- Take a screenshot showing the estimated number of rows in the table.

bytes

Streaming buffer statistics

Estimated size	62.9 MB
Estimated rows	374,849
Earliest entry time	Mar 8, 2025, 11:31:55 AM UTC-8

- Take a screenshot showing the per-minute number of rides, passengers, and revenue for the data collected

Untitled query RUN SAVE DOWNLOAD SHARE SCHEDULE OPEN IN MORE

```

1 SELECT
2   FORMAT_TIMESTAMP("%R", timestamp, "America/Los_Angeles") AS minute,
3   COUNT(DISTINCT ride_id) AS total_rides,
4   SUM(passenger_count) AS total_passengers,
5   SUM(meter_reading) AS total_revenue
6 FROM
7   taxirides.realtime
8 WHERE
9   ride_status = 'dropoff'
10 GROUP BY
11   minute
12 ORDER BY
13   minute ASC

```

Query results

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	minute	total_rides	total_passengers	total_revenue		
1	11:29	11	18	173.4500000000...		
2	11:30	117	206	1825.410007200...		
3	11:31	112	176	1643.7599987		
4	11:32	109	164	1515.8800062		
5	11:33	114	191	1752.440005000...		
6	11:34	109	169	1775.9499981		
7	11:35	107	172	1621.049995599...		

- Take a screenshot showing the plot for your data for your lab notebook

